

2026



Professional Certificate Course in

# **AI-ASSISTED WEB DEVELOPMENT (FRONTEND)**



Barakah Charity  
**IT Center Sri Lanka**



# MODULES

**Course Duration:** 3 Months (48 Hours)

**Course Schedule:** Saturday & Sunday

**Delivery Mode:** On-site with Practical

**Target Audience:** Students after O/L or A/L with basic computer literacy.

**Goal:** Equip beginners with modern web development skills enhanced by Artificial Intelligence tools and automation techniques.



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# COURSE RATIONALE



The rapid digital transformation of businesses has created high demand for skilled web developers who can build responsive, interactive websites efficiently. Modern industry practices increasingly rely on Artificial Intelligence (AI) tools to enhance productivity, code quality, debugging, and deployment. This course is designed to equip learners with foundational web development skills using HTML, CSS, and JavaScript, enhanced through responsible and ethical use of AI-assisted development tools. The program prepares learners for entry-level employment, freelancing, or further NVQ progression in ICT.



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## COURSE OBJECTIVES

The main objectives of this course are to:

- Develop foundational competency in frontend web development using HTML5, CSS3, and JavaScript aligned with industry standards.
- Enable learners to understand web architecture and the role of frontend technologies within modern software systems.
- Introduce AI-assisted development tools to improve coding efficiency, debugging accuracy, and development productivity.
- Train learners to use AI responsibly for code generation, content creation, and design support without over-reliance.
- Build problem-solving and logical thinking skills required for client-side programming.
- Prepare learners to deploy and publish websites using standard hosting and version-control platforms.
- Develop job-ready competencies suitable for junior web developer roles, freelancing, or further NVQ-level ICT studies.
- Promote ethical, secure, and professional practices in web and AI-assisted software development.







## TEACHING & LEARNING METHODS

- Interactive lectures
- Practical / laboratory sessions
- Demonstrations
- Group work and discussions
- Assignments and presentations

## ASSESSMENT METHODS

Learner performance will be assessed using:

- Continuous assessment (quizzes, assignments)
- Practical / skills-based assessments
- Presentations / demonstrations
- Participation and attendance

## ASSESSMENT CRITERIA & WEIGHTING

| Assessment Component                 | Weight (%)  |
|--------------------------------------|-------------|
| Continuous Assessment                | 40%         |
| Practical Assessment % mini Projects | 30%         |
| Final Evaluation / Project           | 30%         |
| <b>Total</b>                         | <b>100%</b> |



## Module 01

### Introduction to Web & AI Concepts

- Web overview (client-server model, browsers, HTTP)
- Frontend & Backend overview
- Introduction to Artificial Intelligence
- Role of AI in software development
- Development environment setup (VS Code or Antigravity, GitHub, ChatGPT, DeepSeek).
- **Mini Project:** Use ChatGPT or Deepseek to plan your first website idea.(demostration)



## Module 02

### HTML5 Development with AI Assistance

- HTML structure and syntax
- Text, images, links, lists, tables
- Forms and input elements
- Semantic HTML5 elements
- AI-assisted HTML template generation.
- **Mini Project:** Create a personal webpage using AI-generated content.



## Module 03

### CSS & Design with AI Tools

- CSS fundamentals and box model
- Colors, typography, layouts
- Flexbox and Grid basics
- Responsive design principles.
- AI design tools (Canva Magic Design)
- **Mini Project:** Design a landing page using an AI design assistant.





## Module 04

### JavaScript Essentials & AI-Assisted Debugging

- JavaScript syntax and logic
- Variables, functions, conditions, loops
- DOM manipulation
- Event handling
- AI-assisted debugging and optimization
- **Mini Project:** Build an AI-Assisted Enterprise level website



## Module 05

### AI-Driven Developer Tools & Prompt Engineering

- GitHub Copilot
- ChatGPT Code Interpreter
- Lovable AI, AI Studio
- AI-generated content and SEO basics
- Prompt engineering fundamentals
- Ethical and responsible AI usage
- **Hands-on:** Generate a Static portfolio site using AI assistance



## Module 06

### Hosting, Deployment & AI-Assisted Publishing

- Web hosting and server basics
- Collaboration and version control tools(Jira, Git, github)
- Domains and DNS
- GitHub Pages, vercel(optional)
- AI-assisted deployment guidance
- Security and responsible publishing





## Module 07

### Final AI-Enhanced Project

- Plan and build a full AI-assisted website (portfolio/blog/e-commerce)
- Integrate one AI feature (chatbot, image generator, content creator).
- **Outcome:** Professional presentation and review for AI Web Development career readiness.

